

Lab Assignment -02

Exploring packet sniffer and packet Analyzer:-

Question 1- What are the different protocols you observe at the following layers of the protocol stack?

Solution- (A) Application Layer

- HTTP
- HTTPS/TLS
- DNS

(B) Transport Layer

- TCP
- UDP

(C) Network Layer

- IPv4

Question 2:- What is the total amount of data being received for the following two cases

(A) When you access <http://neverssl.com/>

(B) When you access <http://www.cornell.edu/>

Solution:-

(A) while accessing http://neverssl.com / the total data received are 2146 bytes

(B) while accessing https://www.cornell.edu the total data received are 17546 bytes

Question 3 - Answer the following when you access the site http://neverssl.com.

(A) How many HTTP GET request do you observe? List down the GET request.

Solution :- There are 3 HTTP GET request

- GET /online HTTP/1.1
- GET /online HTTP/1.1
- GET /favicon.ico HTTP/1.1

(B) What version of HTTP the server is running.

Solution :- HTTP/1.1

(C) For each of the HTTP GET request find out:

1) The total no of TCP segments being received

Solution :- 2 TCP segment.

11) The total amount of data being received in the corresponding HTTP response message.

Solution :- 2146 byte of data.

D) Identify TCP three way handshake used to establish the connection with the <http://neversal.com>. Specify SYN, SYN-ACK, ACK packets and write the sequence no for each.

Solution	packet	sequence no
	SYN	917865950
	SYN-ACK	1836722572 (ACK=917865951)
	ACK	917865951 (ACK=1836722573)